CASE STUDY 4

Advanced Numeric Pattern Generator

Python Activity-Based Learning (ABL)

1. Problem Statement

Write a Python program that accepts a number of rows from the user and prints a number pattern. Each row should begin at 1 and print numbers up to the row number. The program should reject row counts less than 1.

2. Objectives

• To practise input and output statements in Python.

• To use for loops and nested loops.

• To understand how row and column values create a pattern.

• To validate user input.

3. Assumptions

• The user enters a whole number for the number of rows.

• The number of rows must be greater than zero.

• The pattern uses numbers starting from 1 in every row.

4. Algorithm

1. Start the program.

2. Read the number of rows from the user.

3. If rows are less than 1, display an error message.

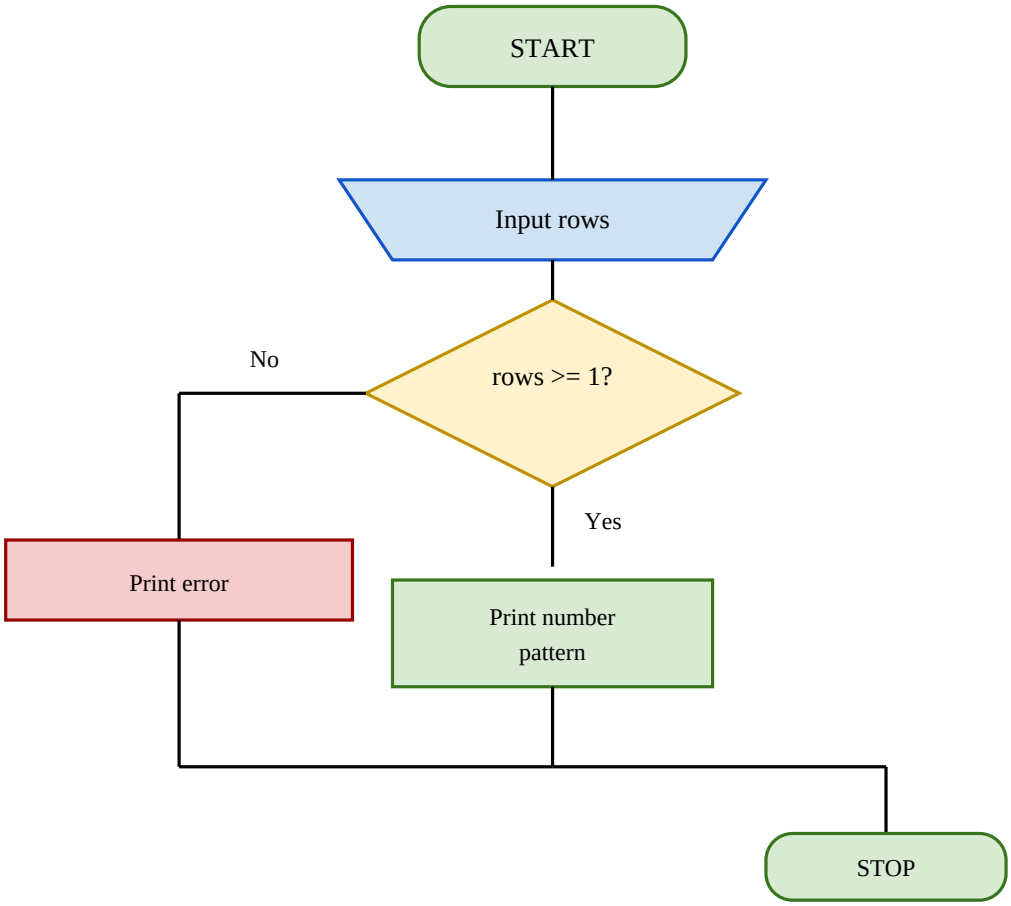
4. Otherwise, use an outer loop from 1 to rows.

5. For each row, use an inner loop from 1 to the row number and print each number.

6. Move to the next line after each row.

7. Stop.

5. Flowchart



6. Pseudocode

```
START
INPUT rows
IF rows < 1 THEN
    PRINT "Please enter a number greater than 0"
ELSE
    FOR i = 1 TO rows
        FOR j = 1 TO i
            PRINT j without a new line
        END FOR
        PRINT a new line
    END FOR
END IF
STOP
```

7. Python Source Code

```
rows = int(input("Enter number of rows: "))

if rows < 1:
    print("Please enter a number greater than 0")
else:
    for i in range(1, rows + 1):
        for j in range(1, i + 1):
            print(j, end=" ")
        print()
```

8. Test Cases

Test	Input	Expected Output
1	5	1\n1 2\n1 2 3\n1 2 3 4\n1 2 3 4 5

Test	Input	Expected Output
2	1	1
3	0	Please enter a number greater than 0
4	-3	Please enter a number greater than 0

9. Sample Output (Output Screenshot Reference)

```
Enter number of rows: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

The above sample output can be used as the expected output when running the program in VS Code. For the final submission, take a screenshot of your own program and output.

10. Conclusion

This program generates a simple number pattern using nested for loops. It helped me understand how loops work together and how to check user input before printing the pattern.